ME 400

Thesis & Project Course

Briefing to Students: How to Write and Present Your Thesis

- What goes in each chapter
- Grading criteria mapped to content
- Oral presentation & defense tips

Sheikh Mohammad Shavik, Ph.D.
Associate Professor
Department of Mechanical Engineering
Bangladesh University of Engineering & Technology (BUET)

"Disclaimer: AI tools were used to assist in preparing certain contents in this presentation. All material has been reviewed and adapted for accuracy and relevance."

How Are You Graded?

Supervisor	External Examiner
Problem Formulation & Literature	Problem Definition & Relevance
Methodology, Analysis & Results	Depth of Literature Review
Modern Engineering Tools	Methodology & Research Design
Health, Safety, Legal, Societal & Cultural Issues	Analysis & Interpretation
Environmental Impact & Sustainability Aspect	Modern Engineering Tools
Ethics & Plagiarism	Environmental Impact & Sustainability Aspect
Teamwork & Individual Contribution	Health, Safety, Legal, Societal & Cultural Issues
Written Communication	Critical & Creative Thinking
Critical Thinking & Learning	Oral Communication & Professionalism
	Response to Questions (Defense)

⚠ **Complex Engineering Problem (CEP) must be addressed — see later section of this guide for details.**

Your Thesis Report — Chapter by Chapter

Each chapter maps directly to grading criteria. Missing a chapter element = lost marks

Ch 1 Introduction

Problem statement, motivation of the work, objectives, significance, literature review

Ch 2 Methodology

Research design, experimental setup or simulation model, tools used

Ch 3 Results & Discussion

Data analysis, graphs, interpretation, engineering reasoning, outcomes of the thesis

Ch 4 Broader Impacts

- Health, safety, legal, ethical, societal and cultural issues and obligation
- Environmental impact and sustainability aspect

Ch 5 Addressing Complex Engineering Problems

Demonstrate how the P1–P7 range of Complex Engineering Problems (CEP) is addressed

Ch 6 Conclusions

Key findings, innovation/novelty, recommendations, future work

References & Appendix

Raw data, code, calibration, full reference list (consistent style)

Chapter
1

Chapter 1: Introduction

1.1 Background & Motivation

- Set real-world context — why does this problem matter?
- State the engineering significance (economic, industrial, societal)
- Identify the knowledge/technology gap that your work addresses

1.2 Problem Statement & Objectives

- One clear, specific problem statement sentence
- List 2–4 measurable objectives (use action verbs: investigate, design, analyse, compare)
- State the scope and limitations of your study

1.3 Literature Review

- Review at least 20–30 peer-reviewed sources (journals > conference papers > textbooks)
- Critically compare approaches — do NOT just summarise each paper
- End with a gap analysis: what is still unknown that your work addresses?
- Tip: use citation manager (Zotero/Mendeley); consistent referencing style (ASME)

1.4 Thesis Organization

- Brief paragraph explaining what each chapter covers
- Optional: include a flow-chart of the study

Grading links → Problem Formulation & literature, Problem definition & relevance, Depth of literature review

Chapter 2

Chapter 2: Methodology & Research Design

2.1 Research Approach

- Is your work experimental, numerical/simulation, analytical, or a combination?
- Justify your choice of approach — why is it suitable for the problem?
- State the overall workflow/flowchart of the methodology

2.2 Experimental Setup (if applicable)

- Describe apparatus: dimensions, make/model, sensor specifications
- State test conditions: range of variables, controlled parameters
- Uncertainty & calibration: how accurate are your measurements? (±%)
- Safety precautions taken during experiments

2.3 Numerical/Simulation Setup (if applicable)

- Governing equations solved (e.g., Navier–Stokes, heat equation, structural equilibrium)
- Software used (ANSYS, MATLAB, OpenFOAM, LAMMPS, Python etc.) + hardware specs
- Boundary conditions, mesh details, and convergence criteria
- Validation: How did you verify your model against known results?

2.4 Use of Engineering Tools, Data Collection & Processing

- Show evidence of tool usage: mesh images, sensor logs
- Justify tool selection — why ANSYS and not OpenFOAM? Why MATLAB and not Excel?
- Demonstrate correct setup: BCs, solver settings, material properties
- How data was recorded, sampled, and stored
- Statistical treatment: averages, standard deviation, error bars

Grading links → Methodology & design, Modern engineering tool usage

Chapter 3

Chapter 3: Results & Discussion

Presenting Results

- Use numbered Figures and Tables with descriptive captions
- Label all axes with quantity + units (e.g., Temperature (°C), Time (sec))
- Include error bars where appropriate; state uncertainty
- Do NOT just dump raw numbers — choose the most meaningful form

Analysis & Interpretation

- Explain WHY the trend occurs using engineering/physics principles
- Compare results with theoretical expectations or published data
- Quantify performance: e.g., '23% improvement over baseline'
- Discuss anomalies honestly — what caused them?

Discussion vs Conclusion — know the difference

- Discussion: interpret and argue what the results mean
- Conclusion: summarise what was definitively proven / achieved

Grading links → Analysis & Results, Analysis & Interpretation

Chapter 4

Chapter 4: Broader Impacts

4.1 Health, Safety, Legal, Social & Cultural Issues [note: all of these issues might not be present in your project] ★must

- What hazards exist in your experiment or application?
- What safety measures or PPE were used?
- Legal/regulatory context: standards (ISO, BNBC, ASME) applicable
- Social/cultural: who benefits or is affected by your work?
- Example: During testing what safety protocols were followed

4.2 Environmental Impact & Sustainability ★must

- Assess energy use, emissions, or material waste in your study
- Does your solution reduce environmental footprint compared to the status quo?
- Discuss lifecycle: manufacturing → use → disposal
- Reference SDGs (Sustainable Development Goals) if relevant
- Example: 'Proposed design reduces power consumption by 18% → lower CO_2

Ethical Responsibilities & Plagiarism

- Similarity percentage — keep below 15%
- Proper citation of all sources, figures, and borrowed ideas
- Data integrity: do not fabricate, falsify or cherry-pick results

Grading links → Environmental impact & sustainability, Health, Safety, Legal, Societal & Cultural Issues, Ethics

Chapter 5

Ch 5: Addressing Complex Engineering Problems - What & Why

All thesis projects should address Complex Engineering Problem. You can map the range of complex engineering problems (discussed in the following slides) and consult with your supervisor to confirm.

What is the CEP Mapping Chapter?

- A dedicated chapter of 1–2 pages in your thesis report ★must
- It explicitly maps your project to the Washington Accord WP characteristics (WP1–WP7)
- For each WP, you state: Does it apply? If yes — where is the evidence in your thesis?
- Think of it as a self-audit: you prove your problem is complex, not just assume it

What Happens If You Skip It?

- Supervisor and External Examiners have no documented basis to confirm that your thesis have addressed CEP
- You will lose some marks under Critical Thinking and Problem Definition criteria
- For accreditation, an unticked or unjustified CEP means the project does not count as a capstone CEP — weakening PO attainment data

Is Your Project a CEP?

CEP = P1 (mandatory) + at least two of P2–P7. After mapping, you can confirm with your supervisor.

P Code	Aspect	What to show in your thesis	Example
P1 ★	Depth of Knowledge (mandatory)	Use engineering knowledge and first-principles theory (one or more of K3, K4, K5, K6, K8) — not just plug-in formulas	Derive Nu from energy balance before comparing to Dittus–Boelter
P2	Conflicting Requirements	Show trade-off between competing objectives	More fins → higher heat transfer BUT higher pressure drop
P3	Depth of Analysis	No obvious solution; required abstract modelling	Choosing optimal fin profile where no closed-form solution exists
P4	Infrequently encountered Issues	Problem not frequently encountered	Cooling of new GaN semiconductor module — no prior data
P5	Beyond Standards	Solution outside existing engineering codes	Non-standard fin geometry not covered by ASME standards
P6	Stakeholder Diversity	Multiple user groups with different needs	Manufacturer (cost), operator (performance), regulator (safety), user
P7	Interdependence	Multiple sub-systems interacting	Thermal + structural + fluid mechanics all coupled

ME400 — CEP Mapping Chapter: Template & Worked Example

How to Write Each WP Entry

P Code & Name

State the P number and its official name from the rubric (e.g., P2 – Conflicting Requirements)

Applicable?

Answer Yes, No, or Partially. Be honest — not all WPs need to apply; Only WP1 is mandatory.

Justification (2–4 sentences)

Explain WHY it applies (or not). Reference specific aspects of your project — not generic statements.

Evidence location

Cite the section and page where the evaluator can verify (e.g., 'Section 3.2, p. 24 — thermal-hydraulic performance trade-off graph').

Mandatory: P1

P1 (Depth of Knowledge) MUST apply. You can justify the first-principles engineering theory in your work, use of engineering design and tools, literature review.

Worked Example — Heat Transfer in Finned Duct Project

P	Apply?	Justification & Evidence Location
P1 ★ Depth of Knowledge	YES	First-principles energy balance used to derive the Nusselt number correlation (WK3). Governing convection-diffusion equations solved analytically before experimental comparison. See Section 2.1 (p. 18).
P2 Conflicting Requirements	YES	Higher fin density improves heat transfer (Nu ↑) but raises pressure drop (f ↑), increasing pumping power. The thermal-hydraulic performance factor $\eta = (Nu/Nu_o)^{(1/3)}$... captures this trade-off. Section 3.3 (p. 28).
P3 Depth of Analysis	YES	No closed-form solution exists for triangular micro-fin geometry in developing flow regime. Abstract modelling using CFD correlation combined with dimensional analysis. Section 2.3 (p. 22).
P4 Infrequent Issues	PARTIAL	Triangular micro-fin profile in rectangular duct under developing flow not previously studied (confirmed by gap analysis in Section 1.4). Standard wavy-fin data exists; micro-fins at this scale do not.
P5 Beyond Standards	NO	Applicable ASME/ISO standards for heat exchanger testing were followed. Problem is within established code boundaries. Not applicable.
P6 Stakeholder Diversity	PARTIAL	Primary stakeholders: electronics manufacturers (thermal management) and HVAC designers. End users and environmental regulators also affected. Addressed in Chapter 4.
P7 Interdependence	YES	Fluid mechanics (flow regime), heat transfer (convection), structural integrity (fin vibration), and material science (copper vs aluminium) are all coupled sub-problems addressed. Section 2.4 (p. 23).

★ P1 is mandatory for all thesis. A CEP is confirmed when P1 applies AND at least two of P2–P7 apply. Supervisor and External Examiners will cross-check your citations against the actual thesis sections.

Chapter 6

Chapter 6: Conclusions & Recommendations

6.1 Conclusions

- Directly answer each objective stated in Chapter 1 — one-to-one correspondence
- Be specific with numbers: 'efficiency improved by 12% at Re=5000'
- State clearly what was proven, NOT just what was done

6.2 Recommendations & Future Work

- What are the limitations of your current study?
- What would you do with more time/resources?
- Propose specific next steps — not just 'further research is needed'

References & Appendix

- Use consistent citation format throughout
- Appendix (optional): raw data tables, extended derivations, code listings, calibration data

Critical & Creative Thinking

- What makes your work novel? How does it differ from prior studies?
- Did you propose an improved method, a new design, or a new insight?
- Show that you questioned assumptions and went beyond the obvious approach
- Tip: write one paragraph starting with 'The key novelty of this work is...'

ME400 — Oral Presentation & Viva Defense

Presentation Structure (10 minutes)

- Slide 1–2: Title + Outline (~20 sec)
- Slide 3–4: Problem definition & Objectives (~1 min 30 sec)
- Slide 5–6: Methodology (~2 min 30 sec)
- Slide 7–9: Results: use your best outcomes (~3 min)
- Slide 10-11: Broader impacts + CEP mapping (~1 min 30 sec)
- Slide 12: Conclusions (~1 min)

Time Management: Practice out loud at least twice before the day. Finish early — never run over.

Slide Design Tips

- No paragraph-sized text blocks
- Use bullet points, and more graphical tools, figures
- Use large, clear graphs with labelled axes
- Highlight the key result on each graph (arrow or call-out box)
- Keep font ≥ 18pt for body text for readability
- Use slide numbers — evaluators will refer to them during Q&A
- Don’t use too dark theme

Q&A / Defense Tips

- Pause, think, then answer — do not rush or guess
- If unsure: “I didn't investigate that specifically, but based on my results.....”
- Know your numbers: memorize key values, boundary conditions, % differences
- Anticipate questions on assumptions, limitations, and why-not-alternatives

Grading links → Oral Communication & Professionalism, Communication, Q&A Defense

Forced Convective Heat Transfer Enhancement Using Extended Surfaces in a Rectangular Duct

How each chapter of report is addressed in this project

Chapter	What this project includes
Ch 1 – Introduction	Background on heat transfer in electronics cooling (industry need). Lit review: 25 papers on fin geometry, Nu correlations. Gap: no study on triangular micro-fins in developing flow.
Ch 2 – Methodology	Experimental rig: 300 mm duct, 5 copper fins (rectangular vs triangular vs wavy). K-type thermocouples, orifice-plate flow meter. Uncertainty: ±1.5%. Energy balance to validate setup.
Ch 3 – Results and Discussions	Nu vs Re graphs (4000–18000). Friction factor vs Re. Thermal-hydraulic performance (η). Found triangular fins 18% higher heat transfer at Re=10000 with 12% higher pressure drop.
Ch 4 – Broader Impacts	Environmental: lower thermal resistance → smaller coolers → less material. Safety: electrical isolation, burn hazard PPE. Society: benefit to HVAC & electronics sector in Bangladesh.
Ch 6 – Conclusion	Novelty: first study to compare these 3 fin shapes under identical duct conditions. Recommends triangular fins for moderate-Re applications. Future: nanofluids, different materials.

For Chapter 5: Addressing CEP of this project, check slide 10

FEA-Based Structural Analysis and Design Optimisation of an Automotive Connecting Rod

How each chapter is addressed in this project

Chapter	What this project includes
Ch 1 – Introduction	Connecting rod failures → engine downtime (industry cost). Lit review: 28 papers on FEA of con-rods, fatigue life, material selection. Gap: no study on composite alternatives for CNG engines.
Ch 2 – Methodology	CAD geometry in SolidWorks. ANSYS Mechanical: static structural + fatigue (Goodman criterion). Mesh convergence study (tetrahedral, ~350k elements). Validation against analytical Hertz contact result.
Ch 3 – Results and Discussion	Von Mises stress contours at peak firing load. Max stress 287 MPa (steel) vs 198 MPa (Ti-6Al-4V). Safety factor maps. Mass reduction: 34% with titanium alloy, 41% with CFRP. Fatigue life > 10^8 cycles for both.
Ch 4 – Broader Impacts	Environmental: lighter con-rod → 3–5% fuel saving → ~200 kg CO_2/year per vehicle. Safety: con-rod burst scenario analysed. Legal: relevant standards. Social: benefit to local auto-parts industry.
Ch 6 – Conclusion	Novelty: first FEA comparison of 4 materials for CNG connecting rods in Bangladeshi engine context. CFRP optimal but cost prohibitive; Ti-6Al-4V recommended near-term. Future: dynamic + thermal coupled analysis.

For Chapter 5: Addressing CEP of this project, check next slide

EXAMPLE PROJECT 2 (Simulation) — Chapter 5: Addressing Complex Engineering Problem (CEP)

Project: FEA-Based Structural Analysis and Design Optimisation of an Automotive Connecting Rod | *Tools: ANSYS Mechanical, SolidWorks* | *Materials: Steel, Al-alloy, Ti-6Al-4V, CFRP*

P Code	Applies?	Justification (What makes it complex)	Evidence Location in Thesis
P1 ★ Depth of Knowledge (mandatory)	YES	First-principles solid mechanics used throughout: constitutive equations of elasticity, von Mises yield criterion, and Goodman fatigue criterion are theoretically derived. FEA based on variational energy principles. CFRP modelled using fibre-mechanics laminate theory, not empirical lookup.	*§2.1 Governing equations (p.15)* *§2.3 Goodman criterion derivation (p.21)* *§2.4 CFRP laminate theory (p.24)*
P2 Conflicting Requirements	YES	Three simultaneous trade-offs: (i) mass reduction (CFRP −41%) vs local manufacturing cost; (ii) Ti-6Al-4V high stiffness vs lower fatigue notch sensitivity vs steel; (iii) lighter con-rod improves fuel economy but reduces damping, increasing vibration risk. No single material satisfies all objectives.	*§3.3 Mass vs cost trade-off table (p.31)* *§3.4 Safety factor vs fatigue life (p.33)* *§5.1 Design recommendation (p.42)*
P3 Depth of Analysis	YES	No closed-form solution exists for fatigue life of composite con-rods under combined axial compression (firing) + tension (inertia) at CNG pressures. Multi-physics chain required: combustion load → static FEA → stress concentration → Goodman fatigue life. Mesh convergence study (h-refinement, ~350k elements) conducted with no established benchmark to compare against.	*§2.2 Mesh convergence study (p.18)* *§2.3 Combined loading formulation (p.20)* *§3.2 Stress concentration factors (p.28)*
P4 Infrequent Issues	YES	Literature review (28 papers) confirmed: no prior FEA study compares all four materials for CNG connecting rods. CNG engines produce higher peak cylinder pressures than petrol, invalidating direct reuse of existing petrol-engine design data. No established reference solution exists for this specific scenario.	*§1.4 Gap analysis (p.12)* *§2.1 CNG pressure profile (p.16)* *Appendix A Literature summary table*
P5 Beyond Standards	PARTIAL	Steel and Al-alloy con-rod design follows ASME B18 / DIN standards. However, no international standard covers CFRP connecting rods, the composite variant lies entirely outside codes. Goodman fatigue analysis for CFRP under CNG-specific loading required first-principles derivation without guidance from existing codes.	*§1.5 Applicable standards review (p.13)* *§2.4 CFRP design outside standard (p.23)* *§5.2 Recommendation for future standardization (p.44)*
P6 Stakeholder Diversity	YES	Four stakeholder groups with conflicting needs: (i) Automotive OEMs for performance & reliability; (ii) CNG vehicle owners for fuel economy & maintenance cost; (iii) Environmental regulators for CO_2 reduction (lighter con-rod → 3–5% fuel saving → ~200 kg CO_2/yr/vehicle); (iv) Local Bangladeshi auto-parts manufacturers for fabricability & affordability.	*§4.2 Stakeholder analysis (p.37)* *§4.3 Environmental impact (p.38)* *§4.4 Socioeconomic impact on local industry (p.39)*
P7 Interdependence of Sub-problems	YES	Five coupled sub-disciplines integrated: (i) Combustion/fluid mechanics (CNG firing pressure input); (ii) Rigid-body dynamics (inertia loading at rated RPM); (iii) Continuum mechanics / FEA (stress field); (iv) Fracture & fatigue mechanics (Goodman life prediction); (v) Materials science (4-material property database). Each sub-problem's output feeds the next, they cannot be solved independently.	*§2.1 Loading from combustion+dynamics (p.15–17)* *§2.3 FEA → fatigue coupling (p.20)* *§3.1 Integrated results workflow diagram (p.26)*

CEP Verdict: P1 ✓ + P2 ✓ + P3 ✓ + P4 ✓ + P5 (Partial) ✓ + P6 ✓ + P7 ✓ → CEP confirmed.

Key Takeaways

What every student must remember:

Every chapter links to marks

Chapters 1–6 directly map to criteria assessed by both your supervisor and the external examiners. Do not skip the broader-impacts and addressing CEP chapter.

Literature review is analytical, not descriptive

Compare and critique papers; end with a gap that your work fills. Citing 10 papers is not the same as reviewing them critically.

Tools must be shown AND justified

State which tools you have used (hardware and software) why you chose those tools and how you validated it

Results need engineering reasoning

'The value increased' earns little. 'The Nusselt number increased by 18% because turbulence intensifies convective mixing' earns full marks.

Innovation is explicitly graded

Write a dedicated paragraph on novelty. How is your work different from, or better than, what came before?

Practice your presentation

Rehearse aloud. Know your key numbers. Prepare for questions on assumptions and limitations.

Good luck with your defense!